PetVest Hub：各图纸间用全局标签连接（I2C_SDA/I2C_SCL/I2S_*/IMU_INT*/GPS_*/USB_DP…）

01_Power

文件：01_power.kicad_sch

02_MCU

文件：02_mcu.kicad_sch

03_IMU_Mux

文件：03_imu_mux.kicad_sch

04_Health

文件：04_health.kicad_sch

05_Audio

文件：05_audio.kicad_sch

06_GNSS_Display

文件：06_gnss_display.kicad_sch

1S锂电 / ESP32-S3 / 6路IMU / I2S音频 / GNSS / 心率

PetVest

Sheet: /

File: petvest-hub.kicad_sch

Title: PetVest 智能宠物服装 · 主控Hub板

Size: A3	Date: 2026-11-07	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 1/7

电源预算(18h续航目标, 1S 4000mAh≈14.8Wh):
ESP32-S3 Wifi间歇上报 ~80mA | GNSS ~40mA | 6×IMU@104Hz ~5mA
OLED ~12mA | 心率(占空比) ~3mA | 音频待机 ~1mA | 电量计/漏 ~1mA
合计~140–150mA → 18h需≥2700mAh; 4000mAh≈25h在线/40h+(休眠策略)

USB-C: CC1/CC2 5.1k下拉声明默认5V供电; D+/D-经USBLC6 ESD后接ESP32-S3原生USB(IO19/20)

TP4056: R3=1.2k → 充电1A; TEMP接地禁用温度检测(电池包自带PCM); CE接VBUS_F=插USB即充

TPS63061: 2.9–4.5V输入→固定3.3V/1.2A; PS/SYNC=GND自动PFM; FB接VOUT

tools/gen_schematic.py 自动生成

PetVest 智能宠物服装

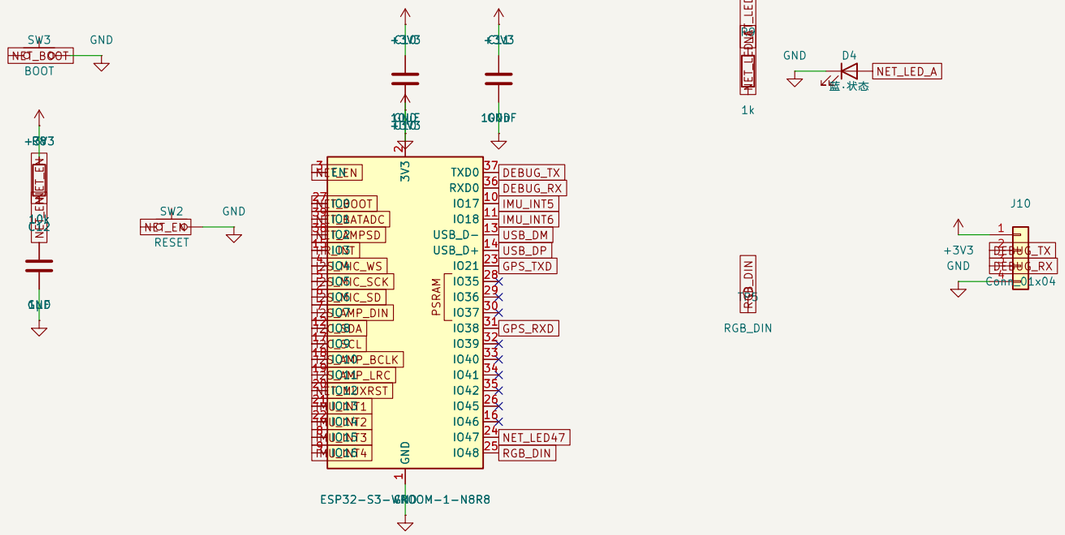
Sheet: /01_Power/
File: 01_power.kicad_sch

Title: 电源: USB-C充电 / 1S锂电 / 3.3V buck-boost / 电量计

Size: A3	Date: 2026-11-07	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 2/7

ESP32-S3-WROOM-1-N8R8 GPIO分配:
  IO0 BOOT键 | IO1 电池分压ADC | IO2 功放SD | IO3 心率INT
  IO4/5/6 I2S麦克风 WS/SCK/SD | IO7/10/11 I2S功放 DIN/BCLK/LRC
  IO8/9 I2C0 SDA/SCL | IO12 复用器RESET | IO13-18 IMU INT1-6
  IO19/20=USB_D-/D+ | IO21→GPS_RX | IO38←GPS_TX | IO47 LED | IO48 RGB扩展
  IO35-37(OctalPSRAM)/IO39-42(仅输入)/IO45-46(boot strap) 悬空NC

□ 天线朝板边; 天线区及两侧≥8mm禁止铺铜走线; 底层勿布高速线

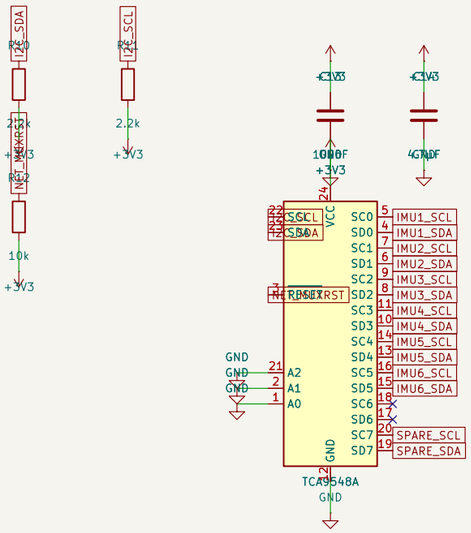


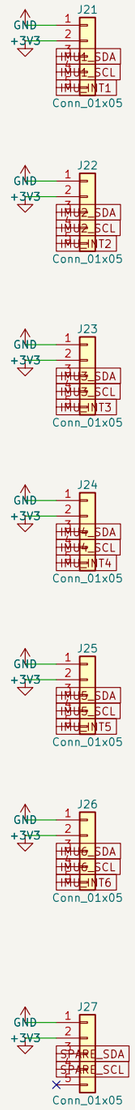
tools/gen_schematic.py 自动生成

PetVest 智能宠物服装

Sheet: /02_MCU/		
File: 02_mcu.kicad_sch		
Title: 主控: ESP32-S3-WROOM-1 (WiFi+BLE+原生USB)		
Size: A3	Date: 2026-11-07	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 3/7

IMU卫星节点 = LSM6DS3TR-C(LGA-14)+2颗电容, 详见 petvest-imu-node 工程
节点I2C地址0x6A(SA0=GND); 上拉电阻仅主控侧2.2k(通道导通时全总线连通)
FPC-5P 1.0mm 线序: 1=3V3 2=GND 3=SDA 4=SCL 5=INT; 软排线≤30cm @400kHz





tools/gen_schematic.py 自动生成

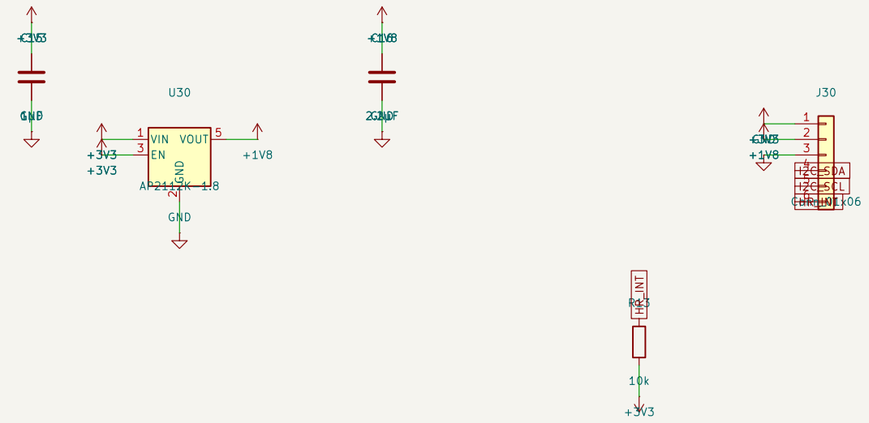
PetVest 智能宠物服装

Sheet: /03_IMU_Mux/
File: 03_imu_mux.kicad_sch

Title: 姿态采集: TCA9548A 8ch I2C复用 → 6×IMU卫星节点

Size: A3	Date: 2026-11-07	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 4/7

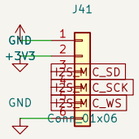
探头卫星板 = MAX30102(OLGA-14) + 100nF/4.7µF, 柔性耳夹或胸贴封装
探头引脚: VDD=1.8V(本板LDO经FPC供电) VLED+=3.3V INT=开漏(本板10k上拉)
□ 犬类毛发遮挡: 反射式PPG建议贴耳廓内侧/大腿内侧, 先验证信号质量再定型

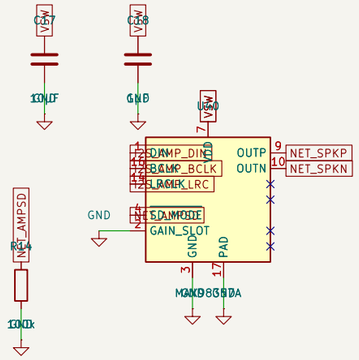


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PetVest 智能宠物服装

Sheet: /04_Health/		
File: 04_health.kicad_sch		
Title: 心率血氧: MAX30102探头卫星 (主板供电/总线)		
Size: A3	Date: 2026-11-07	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 5/7

功放VDD接VSW(开关后电池3.0–4.2V)→音量最大；SD_MODE 100k下拉=默认关断
GAIN_SLOT接地=12dB(量产前按喇叭灵敏度复核)；麦克风L/R接地=左声道
□ 喇叭/麦克风开孔与3D打印外壳声学腔对齐，加防水透声膜



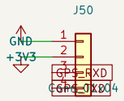


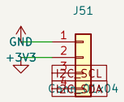


tools/gen_schematic.py 自动生成
PetVest 智能宠物服装

Sheet: /05_Audio/		
File: 05_audio.kicad_sch		
Title: 语音: INMP441 I2S麦克风(卫星) + MAX98357A功放		
Size: A3	Date: 2026-11-07	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 6/7

GNSS: 引脚3=模块TXD→主控IO38(RX); 引脚4=模块RXD←主控IO21(TX); 需3.3V逻辑模块
  天线面朝上, 远离ESP32天线/功放/USB; 犬背上方开窗对天线
OLED: SSD1306 128x64 I2C 0x3C, 挂I2C0直连(不经复用器), 位置=胸带正面





tools/gen_schematic.py 自动生成
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Sheet: /06_GNSS_Display/		
File: 06_gnss_display.kicad_sch		
Title: 定位 GNSS模块 + OLED显示		
Size: A3	Date: 2026-11-07	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 7/7